

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks:30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I R. Rupasree (192H72122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Rupasree

20+20+24+29+14
797
12

Question 1 - Library Late Fine

Knowledge Base

1. If a book is returned late, then a fine is charged.
2. The book was returned late.

Goal- Prove that a fine is charged. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Flight Delay Notification

Knowledge Base

1. If a flight is delayed, then passengers are notified.
2. Flight 202 is delayed.

Goal- Prove that passengers are notified. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Employee Certification

Knowledge Base

1. If an employee completes the training, then the employee receives a certification.
2. Kevin completed the training.

Goal- Prove that Kevin receives a certification. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.

3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Medical Attention Alert

Knowledge Base

1. If a patient has a fever, then the patient needs medical attention.
2. Ravi has a fever.

Goal- Prove that Ravi needs medical attention. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Exam Hall Ticket Eligibility

Knowledge Base

1. If a student completes all assignments, then the student is eligible for the exam.
2. If a student is eligible for the exam, then the student receives a hall ticket.
3. The student completed all assignments.

Goal- Prove that the student receives a hall ticket. using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	20
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	20
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	24
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	19
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	14

① Library Late Fine :-

1) propositional logic :-
Let :

- L = The book was returned late
- F = A fine is charged

Statement :
$$\frac{L \rightarrow F}{L}$$

Goal : F

2) CNF Conversion and Negate Goal :-

Convert Implication: $L \rightarrow F$
 $\neg L \vee F$

clauses are :

$C1 : \neg L \vee F$

$C2 : L$

Negate to Goal : $\neg F$

$C3 : \neg F$

3) Resolution steps :

Resolve $C1$ and $C2$: $(\neg L \vee F), L$

$\therefore F$

Now resolve F with $C3$: $F, \neg F$

Since the empty clause is obtained, the goal is proved.

Therefore, a fine is charged.

⑤ Flight Delay Notification :-

1) propositional logic :-

Let: D = Flight 202 is delayed

N = Passengers are notified

Statement: $D \rightarrow N$
 D

Goal: N

2) CNF conversion and Negate goal :-

Convert Implication: $D \rightarrow N$
 $\neg D \vee N$

clauses: $C_1 : \neg D \vee N$
 $C_2 : D$

Negate the goal: $\neg N$
 $C_3 : \neg N$

3) Resolution steps :-

Resolve C_1 and C_2 : $(\neg D \vee N), D$

Therefore: N

Now resolve N with C_3 :

$N, \neg N$

Since the empty clause is derived, the goal is proved.

Therefore, passengers notified.

2) Employee Certification:-

a) propositional logic:

Let : T = kevin complete the training
 C = kevin receives a certificate

Statement : $T \rightarrow C$
 T

Goal : C

b) CNF conversion and Negate Goal:

Convert Implication:

$$\begin{aligned} T &\rightarrow C \\ \neg T \vee C \end{aligned}$$

clauses:

$$C1: \neg T \vee C$$

$$C2: T$$

Negate the goal : $\neg C$

$$C3: \neg C$$

c) Resolution Steps:

Resolve $C1$ and $C2$,
 $(\neg T \vee C), T$

Therefore: C

Now resolve C with $C3$:
 $C, \neg C$

The Empty clause $\varnothing$ is obtained

Therefore, kevin receives a certificate.

④ Medical Attention Alert

1) propositional logic:

Let: F = Ravi has a fever

M = Ravi needs medical attention

Statements:

$$F \rightarrow M$$

$$F$$

Goal: M

2) CNF conversion and Negate Goal:-

Convert Implication:

$$F \rightarrow M$$

$$\neg F \vee M$$

clauses:

$$C1: \neg F \vee M$$

$$C2: F$$

Negate the goal: $\neg M$

$$C3: \neg M$$

3) Resolution steps:-

Resolve $C1$ and $C2$:

$$(\neg F \vee M), F$$

Therefore: M

Now resolve M with $C3$:

$$M, \neg M$$

The Empty clause is obtained.

Therefore, Ravi needs a medical attention.

5) Exam Hall Ticket Eligibility :-

1) propositional logic :-

Let: A = student completed all assignments
 E = student is eligible for exam
 H = student receives a hall ticket

Knowledge Base:

Statement 1: $A \rightarrow E$

Statement 2: $E \rightarrow H$

Statement 3: A

Goal: H

2) Convert CNF (All implications) :-

Statement 1: Statement 2:

$A \rightarrow E$

$E \rightarrow H$

$[P \rightarrow Q \equiv \neg P \vee Q]$

$\neg E \vee H \rightarrow C2$

$\therefore \neg A \vee E \rightarrow C1$

Statement 3:

$A \rightarrow C3$

$C1: \neg A \vee E$

$C2: \neg E \vee H$

$C3: H$

Negote :- Goal: H
 $\downarrow$
 $\neg H$

3) Resolution :-

Step 1:-

$(\neg A \vee E), A \Rightarrow E$

Step 2:

$$(\neg E \vee H), E \Rightarrow H$$

Step 3:

$$H, \neg H$$

Since the empty clause is obtained, the goal is proved.

$\therefore$ Student receives Hall ticket.

